Solstice® 452A

000000022311

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1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product information

Trade name	:	Solstice® 452A
Number	:	000000022311
Recommended use of the chemical and restrictions on use	:	Refrigerant
Manufacturer or supplier's details	:	Solstice Trading (Shanghai) Co., Ltd. No.69 Xi Ya Rd, Shanghai, 31 200131 China
For further information, please contact:	:	800-522-8001 +1-973-455-6300 (Monday-Friday, 9:00am-5:00pm)
In case of emergency call	:	**Medical: 1 866-479-8788 or +1 303-739-1378**
	:	**Transportation (CHEMTREC): 1-800-424-9300 or +1-703-527-3887** **In Thailand: 1800014808**

2. HAZARDS IDENTIFICATION

Classification of the substance or mixture

Classification of the substance or mixture	:	Gases under pressure, Liquefied gas Short-term (acute) aquatic hazard, Category 3

GHS Label elements, including precautionary statements

Symbol(s) :



Signal word : Warning

Hazard statements : Contains gas under pressure; may explode if heated.
Harmful to aquatic life.

Precautionary statements : **Prevention:**
Avoid release to the environment.

Storage:
Protect from sunlight. Store in a well-ventilated place.

Disposal:
Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification : Excessive exposure may cause central nervous system effects including drowsiness and dizziness. Excessive exposure may also cause cardiac arrhythmia.
Causes asphyxiation in high concentrations. The victim will not realize that he/she is suffocating.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical nature : Mixture

Chemical name	CAS-No.	Concentration
Pentafluoroethane Pentafluoroethane	354-33-6	59.00 %
2,3,3,3-Tetrafluoroprop-1-ene 2,3,3,3-Tetrafluoroprop-1-ene	754-12-1	30.00 %

Difluoromethane Difluoromethane	75-10-5	11.00 %

4. FIRST AID MEASURES

General advice : First aider needs to protect himself.
Move out of dangerous area.
Take off all contaminated clothing immediately.

Inhalation : Move to fresh air.
If breathing is irregular or stopped, administer artificial respiration.
Use oxygen as required, provided a qualified operator is present.
Call a physician.
Do not give drugs from adrenaline-ephedrine group.

Skin contact : After contact with skin, wash immediately with plenty of water.
If there is evidence of frostbite, bathe (do not rub) with lukewarm (not hot) water. If water is not available, cover with a clean, soft cloth or similar covering.
If symptoms persist, call a physician.

Eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
In case of frostbite water should be lukewarm, not hot.
If symptoms persist, call a physician.

Ingestion : As this product is a gas, refer to the inhalation section.
Do not induce vomiting without medical advice.
Call a physician immediately.

Notes to physician : Because of the possible disturbances of cardiac rhythm, catecholamine drugs, such as epinephrine, should be used with special caution and only in situations of emergency life support. Treatment of overexposure should be directed at the control of symptoms and the clinical conditions.

Treat frost-bitten areas as needed.

5. FIREFIGHTING MEASURES

Suitable extinguishing media	:	The product is not flammable. ASHRAE 34 Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Specific hazards during firefighting	:	Contents under pressure. This product is not flammable at ambient temperatures and atmospheric pressure. However, this material can ignite when mixed with air under pressure and exposed to strong ignition sources. Container may rupture on heating. Cool closed containers exposed to fire with water spray. Do not allow run-off from fire fighting to enter drains or water courses. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Fire may cause evolution of: Halogenated compounds Hydrogen fluoride Carbonyl halides Carbon oxides
Special protective equipment for firefighters	:	In the event of fire and/or explosion do not breathe fumes. Wear self-contained breathing apparatus and protective suit. No unprotected exposed skin areas.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Immediately evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Wear personal protective equipment. Unprotected persons

must be kept away.
Remove all sources of ignition.
Avoid skin contact with leaking liquid (danger of frostbite).
Ventilate the area.
After release, disperses into the air.
Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing.
Avoid accumulation of vapours in low areas.
Unprotected personnel should not return until air has been tested and determined safe.
Ensure that the oxygen content is >= 19.5%.

Environmental precautions : Prevent further leakage or spillage if safe to do so.
The product evaporates readily.

Methods and materials for containment and cleaning up : Ventilate the area.

7. HANDLING AND STORAGE

Handling

Precautions for safe handling : Handle with care.
Avoid inhalation of vapour or mist.
Do not get in eyes, on skin, or on clothing.
Wear personal protective equipment.
Use only in well-ventilated areas.
Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50 °C.
Follow all standard safety precautions for handling and use of compressed gas cylinders.
Use authorized cylinders only.
Protect cylinders from physical damage.
Do not puncture or drop cylinders, or expose them to open flame or excessive heat.
Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material.
Do not remove screw cap until immediately ready for use.
Always replace cap after use.

Advice on protection against fire and explosion	:	The product is not flammable. Can form a combustible mixture with air at pressures above atmospheric pressure. Normal measures for preventive fire protection.

Storage

Conditions for safe storage, including any incompatibilities	:	Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use. Keep containers tightly closed in a dry, cool and well-ventilated place. Storage rooms must be properly ventilated. Ensure adequate ventilation, especially in confined areas. Protect cylinders from physical damage.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis
Pentafluoroethane Pentafluoroethane	354-33-6	TWA : Time weighted average	(1,000 ppm)		Solstice:Limit established by Solstice Advanced Materials
		TWA : Time weighted average	4,900 mg/m3 (1,000 ppm)	2007	WEEL:US Workplace Environmental Exposure Level

2,3,3,3-Tetrafluoroprop-1-ene 2,3,3,3-Tetrafluoroprop-1-ene	754-12-1	TWA : Time weighted average	(500 ppm)	03 15 2010	Solstice:Limit established by Solstice Advanced Materials
		STEL : Short term exposure limit	(1,500 ppm)	03 15 2010	Solstice:Limit established by Solstice Advanced Materials
		TWA : Time weighted average	(500 ppm)	2009	WEEL:US Workplace Environmental Exposure Level
Difluoromethane Difluoromethane	75-10-5	TWA : Time weighted average	(1,000 ppm)	1994	Solstice:Limit established by Solstice Advanced Materials
		TWA : Time weighted average	2,200 mg/m3 (1,000 ppm)	2007	WEEL:US Workplace Environmental Exposure Level

Appropriate engineering controls

General room ventilation is adequate for storage and handling.
Perform filling operations only at stations with exhaust ventilation facilities.

Individual protection measures, such as personal protective equipment (PPE)

Respiratory protection	:	In case of insufficient ventilation, wear suitable respiratory equipment. Wear a positive-pressure supplied-air respirator. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. For rescue and maintenance work in storage tanks use self-contained breathing apparatus.
Eye protection	:	Wear as appropriate: Safety glasses with side-shields If splashes are likely to occur, wear: Goggles or face shield, giving complete protection to eyes
Skin and body protection	:	Wear suitable protective equipment. Wear as appropriate: Protective suit
Hygiene measures	:	Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation, especially in confined areas. Avoid contact with skin, eyes and clothing. Remove and wash contaminated clothing before re-use. Keep working clothes separately. When using, do not eat, drink or smoke.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	Liquefied gas
Colour	:	clear colourless
Odour	:	slight ether-like
Odour Threshold	:	Note: No data available

pH	:	Note: No data available
Melting point/ range	:	Note: No data available
Boiling point/boiling range	:	Note: No data available
Flash point	:	Note: Not applicable
Flammability	:	does not ignite
Lower flammability limit	:	Note: Not applicable
Upper flammability limit	:	Note: Not applicable
Vapour pressure	:	1.07 MPa at 21.1 °C(70.0 °F)
Vapour density	:	Note: No data available
Density	:	1.14 g/cm3
Water solubility	:	Note: negligible
Partition coefficient: n-octanol/water	:	Note: No data available
Ignition temperature	:	Note: No data available
Decomposition temperature	:	Note: No data available

10. STABILITY AND REACTIVITY

Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Hazardous polymerisation does not occur.
Conditions to avoid	:	Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50 °C. Decomposes under high temperature. Some risk may be expected of corrosive and toxic decomposition products. Can form a combustible mixture with air at pressures above atmospheric pressure. Do not mix with oxygen or air above atmospheric pressure.
Incompatible materials to avoid	:	Alkali metals Oxidizers (e.g. peroxide residues present in insufficiently cured rubbers) Finely divided metal powders such as aluminum, magnesium, or zinc.
Hazardous decomposition products	:	In case of fire hazardous decomposition products may be produced such as: Halogenated compounds Hydrogen fluoride Carbonyl halides Carbon monoxide Carbon dioxide (CO2)

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure	:	Inhalation, Eye contact, Skin contact, Ingestion
Acute oral toxicity	:	Note: No data available
Acute inhalation toxicity	:	Note: No data available
Acute dermal toxicity	:	Note: No data available

Skin corrosion/irritation	:	Note: No data available
Serious eye damage/eye irritation	:	Note: No data available
Skin sensitisation	:	Note: No data available
Respiratory sensitisation	:	Note: No data available
Repeated dose toxicity		
Pentafluoroethane	:	Species: Rat Application Route: Inhalation Exposure time: (4 Weeks) NOEL: 50000 ppm Subchronic toxicity
2,3,3,3-Tetrafluoroprop-1-ene	:	Species: Rat Application Route: Inhalation Exposure time: (2 Weeks) No-observed-effect level: 50000 ppm Method: OECD Test Guideline 412 Species: Rat Application Route: Inhalation Exposure time: (4 Weeks) NOAEL (No observed adverse effect level): 50000 ppm Method: OECD Test Guideline 412 Species: Rat Application Route: Inhalation Exposure time: (13 Weeks) NOAEL (No observed adverse effect level): 50000 ppm Method: OECD Test Guideline 413 Species: Rabbit, male Application Route: Inhalation

Exposure time: (28 d)
No-observed-effect level: 500 ppm
Method: OECD Test Guideline 412
There are no observed toxicological effects, which result in classification as a specific target organ toxicant.

Species: Rabbit, female
Application Route: Inhalation
Exposure time: (28 d)
No-observed-effect level: 1000 ppm
Method: OECD Test Guideline 412
There are no observed toxicological effects, which result in classification as a specific target organ toxicant.

Species: Mini-pig
Application Route: Inhalation
Exposure time: (28 d)
NOAEL (No observed adverse effect level): 10000 ppm
highest exposure tested

Difluoromethane : Species: Rat
Application Route: Inhalation
Exposure time: (90 d)
NOEL: 50000 ppm
Subchronic toxicity

Germ cell mutagenicity : Note: No data available

Germ cell mutagenicity
2,3,3,3-Tetrafluoroprop-1-ene : Test Method: Unscheduled DNA synthesis
Dose: up to 50,000 ppm (4 weeks)
Method: OECD Test Guideline 486
Result: negative

: Species: Rat
Cell type: Micronucleus
Dose: up to 50,000 ppm (4 weeks)

Method: OECD Test Guideline 474
Result: negative

Difluoromethane : Species: Mouse
Cell type: Bone marrow
Method: Mutagenicity (micronucleus test)
Result: negative

Carcinogenicity
2,3,3,3-Tetrafluoroprop-1-ene : Species: Rat
Note: Not classified as a human carcinogen. Substance not expected to be a carcinogen based on available data.

Reproductive toxicity :
Note: No data available

Teratogenicity
Pentafluoroethane : Species: Rabbit
Application Route: Inhalation exposure
NOAEL,Teratog: 50,000 ppm
NOAEL,Maternal: 50,000 ppm
Note: Did not show teratogenic effects in animal experiments.

Species: Rat
Application Route: Inhalation exposure
NOAEL,Teratog: 50,000 ppm
NOAEL,Maternal: 50,000 ppm
Note: Did not show teratogenic effects in animal experiments.

2,3,3,3-Tetrafluoroprop-1-ene : Species: Rat
Dose: NOAEL (No observed adverse effect level) - 50,000 ppm

Species: Rabbit
Dose: NOAEL (No observed adverse effect level) - 4,000 ppm

Difluoromethane	:	Species: Rat Dose: NOEL - 50,000 ppm Note: Did not show teratogenic effects in animal experiments. Species: Rabbit Dose: NOEL - 50,000 ppm Note: Did not show teratogenic effects in animal experiments.
Specific target organ toxicity - single exposure	:	Note :No data available
Specific target organ toxicity - repeated exposure	:	Note :No data available
Further information	:	Note: Rapid evaporation of the liquid may cause frostbite. Vapours are heavier than air and can cause suffocation by reducing oxygen available for breathing. Handle in accordance with good industrial hygiene and safety practice.

12. ECOLOGICAL INFORMATION

Aquatic and ecotoxicity

Toxicity to fish	:	Note: No data available
Toxicity to daphnia and other aquatic invertebrates	:	Note: No data available
Toxicity to algae	:	Note: No data available

Persistence and degradability

Biodegradability	:	Note: No data available

Bioaccumulative potential

Bioaccumulation	:	Note: No data available

Mobility in soil

Mobility	:	Note: No data available

13. DISPOSAL CONSIDERATIONS

Disposal methods	:	In accordance with local and national regulations.

14. TRANSPORT INFORMATION

ADR

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (PENTAFLUOROETHANE, R-1234yf, DIFLUOROMETHANE)
Class	:	2
Classification Code	:	2A
Hazard Identification Number	:	20
Labels	:	2.2

IATA

UN/ID No.	:	UN 3163
Description of the goods	:	Liquefied gas, n.o.s. (Pentafluoroethane, R-1234yf, Difluoromethane)
Class	:	2.2
Labels	:	2.2
Packing instruction (cargo aircraft)	:	200
Packing instruction	:	200

(passenger aircraft)

IMDG

UN/ID No.	:	UN 3163
Description of the goods	:	LIQUEFIED GAS, N.O.S. (PENTAFLUOROETHANE, R-1234yf, DIFLUOROMETHANE)
Class	:	2.2
Labels	:	2.2
EmS Number 1	:	F-C
EmS Number 2	:	S-V
Marine pollutant	:	no

15. REGULATORY INFORMATION

National regulatory information

: No information available.

Other international regulations

Notification status

USA. List of Active Substances on the Toxic Substances Control Act (TSCA) Chemical Substances Inventory, as amended	:	All substances listed as active on the TSCA inventory
Australian Inventory of Industrial Chemicals	:	All components are listed on the inventory, regulatory obligations/restrictions apply

Canada. Domestic Substances List (DSL), as amended	:	All components of this product are on the Canadian DSL
Japan. Kashin-Hou Law List	:	On the inventory, or in compliance with the inventory
Korea. Existing Chemicals Inventory (KECI)	:	On the inventory, or in compliance with the inventory
Philippines. Inventory of Chemicals and Chemical Substances (PICCS)	:	On the inventory, or in compliance with the inventory
China. Inventory of Existing Chemical Substances (IECSC)	:	On the inventory, or in compliance with the inventory
Thailand. Existing Chemicals Inventory from FDA (TECI List)	:	On the inventory, or in compliance with the inventory
Taiwan Chemical Substance Inventory (TCSI)	:	On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

		HMIS III	NFPA
Health hazard	:	1	2
Flammability	:	1	1
Physical Hazard	:	1	
Instability	:		0

Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.
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Person who prepared the SDS Solstice Product Stewardship Group